Supersymmetric AdS solions and phase transitions in $\mathcal{N} = 4$ SYM

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Outline

1. Motivations
2. AdS Solitons
3. The setup: $U(1)^3$ gauged sugra $D = 5$
4. New soliton solution
5. Holographic renormalization and phase diagram
6. Lift to 10 dimensions

Deformations of $\mathrm{AdS}_5 \times \mathrm{S}^5$

The are many ways to deform $\mathrm{AdS}_5 \times S^5$ vacuum in the literature.

- Witten model ['98 Witten]: Interpretated as dual to $\mathcal{N} = 4$ SYM at finite temperature.
- Continuous distributions of D3-branes: Coulomb Branch ['98 Freedman, Gubser, Pilch, Warner]: Obtained by adding scalar operator of dimension $\Delta = 2$ leading to two possible manifolds manifolds $\mathscr{M}_\pm$ described by a dimensionless parameter $\pm\ell^2/L^2$.

$$\mathrm{d}s_{10}^2 = H^{-1/2}(-\mathrm{d}t^2 + \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2) + H^{1/2}\sum_{i=1}^{6}\mathrm{d}y_i^6$$

$$H(\vec{y}) = \int_{|\vec{w}|\leq\ell} \mathrm{d}^n\vec{w}\sigma_n(\vec{w})\frac{L^4}{|\vec{y}-\vec{w}|}, \quad \begin{array}{l} \sigma_2(\vec{w}) = \frac{1}{\pi\ell^2}\theta(\ell^2 - \vec{w}^2) \\ \sigma_4(\vec{w}) = \frac{1}{\pi^2\ell^2}\delta^{(4)}(\ell^2 - \vec{w}^2) \end{array}$$

Continuous distributions of D3-branes

The $n = 2$ ($\epsilon = 1$) and $n = 4$ ($\epsilon = -1$) continuous distribution has the metric

$$\mathrm{d}s_{10}^2 = \frac{\zeta(r,\theta) r^2}{L^2}\left(\mathrm{d}x_{1,3}^2 + \frac{L^4}{r^4}\frac{\mathrm{d}r^2}{\lambda(r)^6}\right) + \frac{L^2}{\zeta(r,\theta)}(\zeta(r,\theta)^2 \mathrm{d}\theta^2 + \cos^2\theta \mathrm{d}\Omega_3^2 + \lambda(r)^6 \sin^2\theta \mathrm{d}\Omega_1^2)$$

where

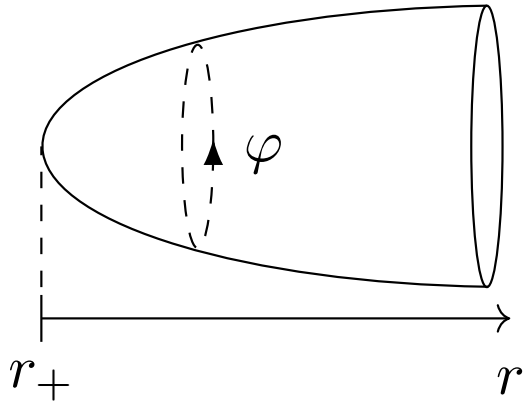
$$\zeta(r,\theta) = 1 + \epsilon\frac{\ell^2}{r^2}\cos^2\theta\,, \quad \lambda(r)^6 = 1 + \epsilon\frac{\ell^2}{r^2}\,.$$

Symmetry:

$$SO(4) \times SO(2)$$

How we will deform it?

- In the asymptotically flat spaces, Witten showed that the Kaluza-Klein soliton with anti-periodic boundary conditions decay into a bubble of nothing.
- The AdS soliton ['98 Horowitz,Myers]
- SUSY AdS solitons D=4,D=5 ['21 Anabalon-Ross]
- SUSY asymptotically flat D=4 soliton ['21 Canfora,Oliva,MO] and soliton D5 configuration ['22,'23 Nuñez,Stuardo,MO].
- SUSY hairy AdS soliton D=4 ['22 Anabalon-Ross-Trigiante-Gallerati]
- Recently, AdS solition=smooth distribution of M2-branes ['24 Anabalon-Astefanesei-Gallerati-Oliva]



Review on AdS solitons

Horowitz-Myers AdS soliton

Take Schwarzshild AdS

$$\mathrm{d}s^2 = -\left(\frac{r^2}{\ell^2} - \frac{r_+^3}{\ell^2 r}\right)\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{\frac{r^2}{\ell^2} - \frac{r_+^3}{\ell^2 r}} + r^2(\mathrm{d}x^2 + \mathrm{d}y^2)$$

Conisdering a double Wick rotation on

$$t \to i\varphi\,, \quad x \to it\,,$$

we get the Horowitz-Myers AdS soltion ['98 Horowitz,Myers]

$$\mathrm{d}s^2 = -r^2\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{\frac{r^2}{\ell^2} - \frac{r_+^3}{\ell^2 r}} + \frac{1}{\ell^2}\left(r^2 - \frac{r_+^3}{r}\right)\mathrm{d}\varphi^2 + r^2\mathrm{d}y^2\,,$$

$$\delta = \frac{4\pi\ell^2}{3r_+}\,, \quad \mathrm{d}s^2 \overset{r\sim\infty}{\longrightarrow} r^2(-\mathrm{d}t^2 + \mathrm{d}\varphi^2 + \mathrm{d}y^2) + \frac{\mathrm{d}r^2}{r^2}$$

non-susy, $\phi \sim \phi + \delta$

Anabalon-Ross soliton

The action of $\mathcal{N}=2$ gauged supergravity in $D=4$ is

$$S=\frac{1}{2}\int \mathrm{d}^4x\sqrt{-g}\left(R-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}+\frac{3}{2\ell^2}\right)+\text{fermions}$$

Charged planar Schwarzschild-AdS is a solution

$$\mathrm{d}s^2=-\left(\frac{r^2}{\ell^2}-\frac{\mu}{r}+\frac{Q^2}{r^2}\right)\mathrm{d}t^2+\frac{\mathrm{d}r^2}{\frac{r^2}{\ell^2}-\frac{\mu}{r}+\frac{Q^2}{r^2}}+r^2\left(\mathrm{d}x^2+\mathrm{d}y^2\right)\ ,$$

$$A=\left(\frac{2Q}{r}-\frac{2Q}{r_0}\right)\mathrm{d}t\,,$$

Anabalon-Ross soliton

Considering the following analytic continuation on RN

$$t \to i\varphi\,, \qquad x \to it\,, \qquad Q \to -iQ\,,$$

leads to the Anabalon-Ross solition $r \in [r_+, \infty[, \varphi \in [0, \delta]$

$$\mathrm{d}s^2 = -r^2\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{\frac{r^2}{\ell^2} - \frac{\mu}{r} - \frac{Q^2}{r^2}} + \left(\frac{r^2}{\ell^2} - \frac{\mu}{r} - \frac{Q^2}{r^2}\right)\mathrm{d}\varphi^2 + r^2\mathrm{d}y^2\,,$$

$$A = \left(\frac{2Q}{r} - \frac{2Q}{r_0}\right)\mathrm{d}\varphi\,, \qquad \delta = \frac{4\pi\ell^2 r_0^3}{3r_0^4 + Q^2\ell^2}\,.$$

- Uniparametric family of magnetic regular configuration.
- Supersymmetric when $\mu = 0$
- Asymptotically locally AdS
- The energy and current in the dual theory are

$$\langle T_{tt}\rangle = -\frac{\mu}{2\ell^2}\,, \qquad \langle J_\varphi\rangle = 2Q$$

Type IIB

Starting from type IIB sugra in the metric-F_5 sector

$$S = \int_{\mathscr{M}_{10}} R \star 1 - \frac{1}{4} F_5 \wedge \star F_5 + \ldots (\Phi, H_3, F_1, F_3)$$

the fermions are one gravitino Ψ_μ and one dilatino λ. Setting (consistently) $\Phi = H_3 = F_1 = F_3 = 0$ we get the following SUSY variations

$$\delta \Psi_\mu \mathrm{d}x^\mu = \mathrm{d}\epsilon + \frac{1}{4}\omega_{ab}\Gamma^{ab}\epsilon + \frac{1}{16}\not{F}_5 i\sigma_2 \Gamma_a e^a \epsilon \,,$$
$$\delta\lambda = 0 \,.$$

$\not{F}_p \equiv \frac{1}{p!} F_{a_1 \ldots a_p} \Gamma^{a_1 \ldots a_p}$

Kaluza-Klein on S^5

The KK ansatz is ['99 Cvetic,Duff,Hoxha,Liu,Lu,Lu,et al] which will leads to a gauged $\mathcal{N} = 2$ is of the form

$$\mathscr{M}_{10} = \mathscr{M}_5 \ltimes \tilde{S}^5$$

We use the following coordinates for the round S^5

$$\mathrm{d}s^2_{S^5_{\text{round}}} = \sum_{i=1}^{3} \mathrm{d}\mu_i^2 + \mu_i^2 \mathrm{d}\phi_i^2 \,,$$

where $\sum_i \mu_i^2 = 1$. The five dimensional theory will contain the following fields

- 3 vectors, 2 scalars, 2 gravitino and 2 dilatino

Kaluza-Klein on S^5

Defining $\tilde{\Delta} = \sum_{i=1}^{3} \mu_i^2 X_i, \quad \sum_{i=1}^{3} \mu_i^2 = 1$

$$X_i = e^{-\frac{1}{2}\vec{a}_i \cdot \vec{\Phi}}, \qquad \vec{\Phi} = (\Phi_1, \Phi_2)$$

$$\vec{a}_1 = (\frac{2}{\sqrt{6}}, \sqrt{2}), \quad \vec{a}_2 = (\frac{2}{\sqrt{6}}, -\sqrt{2}), \quad \vec{a}_3 = (-\frac{4}{\sqrt{6}}, 0).$$

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The reduction is

$$\mathrm{d}s_{10}^2 = \tilde{\Delta}^{1/2} \mathrm{d}s_5^2 + L^2 \tilde{\Delta}^{-1/2} \sum_{i=1}^{3} X_i^{-1} \left[\mathrm{d}\mu_i^2 + \mu_i^2 (\mathrm{d}\phi_i + L^{-1} A^i)^2 \right],$$

Kaluza-Klein on S^5

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$$F_5 = G_5 + \star G_5,$$

$$G_5 \equiv \frac{2}{L} \epsilon_5 \sum_{i=1}^{3} (X_i^2 \mu_i^2 - \tilde{\Delta} X_i) - \frac{L}{2} X_i^{-1} \star_5 \mathrm{d}X_i \wedge \mathrm{d}\mu_i^2$$

$$+ L^2 \sum_i X_i^{-2} \mu_i \mathrm{d}\mu_i \wedge (\mathrm{d}\phi_i + L^{-1} A^i) \wedge \star_5 F^i$$

$\mathcal{N} = 2$ gauged sugra $D = 5$

The resulting theory in $D = 5$ has the following action principle

$$S = \frac{1}{2\kappa} \int (R - \sum_{i=1}^{3} 4L^{-2} X_i^{-1}) \star_5 1 - \frac{1}{2} \mathrm{d}\Phi_1 \wedge \star \mathrm{d}\Phi_1 - \frac{1}{2} \mathrm{d}\Phi_2 \wedge \star \mathrm{d}\Phi_2$$
$$- \frac{1}{2} \sum_{i=1}^{3} X_i^{-2} F^i \wedge \star_5 F^i + A^1 \wedge F^2 \wedge F^3 \,.$$

- $\mathcal{N} = 2$ gauged sugra. The fermions are 2 dilatinos λ_1, λ_2 and 2 gravitino ψ_μ^a
- The local spinorial parameter generating the SUSY transformations is a symplectic Majorana ϵ^a, $a = 1, 2$. We define a complex spinor which

$$\hat{\epsilon} = \epsilon^1 + \mathrm{i}\epsilon^2 \,,$$

useful to integrate the Killing spinor equations

$\mathcal{N} = 2$ gauged sugra $D = 5$

The susy variation in terms of $\hat{\epsilon}$ of the gravitino is [2000 Klemm,Sabra].

$$\delta\hat{\psi}_\mu \mathrm{d}x^\mu = \mathrm{d}\hat{\epsilon} + \mathbf{W}\hat{\epsilon}\,,$$

$$\mathbf{W} \equiv \frac{1}{4}\omega_{ab}\gamma^{ab} + L^{-1}\sum_i(-\frac{\mathrm{i}}{2}A^i + \frac{1}{3!}\gamma_c e^c X_i)$$

$$+ \frac{\mathrm{i}}{4!}(\gamma_c\gamma^{ab} - 6\delta^a_c\gamma^b)e^c\sum_i X_i^{-1}F^i_{ab}$$

While the variation of the dilatinos are

$$\delta\lambda_1 = \sum_{i=1}^{3}\mathbf{\Omega}_i\frac{\partial X_i}{\partial\Phi_1}\hat{\epsilon}\,, \qquad \delta\lambda_2 = \sum_{i=1}^{3}\mathbf{\Omega}_i\frac{\partial X_i}{\partial\Phi_2}\hat{\epsilon}$$

$$\mathbf{\Omega}_i \equiv -\frac{1}{8}X_i^{-2}\gamma^{ab}F^i_{ab} - \frac{\mathrm{i}}{4}X_i^{-2}\left(\frac{\partial X_i}{\partial\Phi_1}\not{\partial}\Phi_1 + \frac{\partial X_i}{\partial\Phi_2}\not{\partial}\Phi_2\right)\mathbb{1} + \frac{\mathrm{i}}{2L}\mathbb{1}$$

Let us move the solution...

New soliton solution

We generalize the Anabalon-Ross solution including a non trivial VEV and non trivial currents in $D = 5$.
Seed spacetime:

Seed spacetime

Particular case of static hairy electrically charged black hole.
The lift of this correspond to a spinning D3 brane.
['99 Cvetic,Duff,Liu,Lu,Lu,Pope,et al]

New soliton solution

Considering coordinates $[t, x, \phi, y, z]$

$$\mathrm{d}s_5^2 = \Omega(x)\left(-\mathrm{d}t^2 + \frac{\mathrm{d}x^2}{4x^2(x-1)F(x)\eta} + F(x)\mathrm{d}\phi^2 + \mathrm{d}y^2 + \mathrm{d}z^2\right)$$

$$\Omega(x) = \frac{x^{2/3}\eta}{x-1}\,, \qquad F(x) = 1 + \frac{(1-x)^2(q_1^2 - q_2^2 x)}{\eta x^2}\,.$$

The matter fields

$$\Phi_1 = \sqrt{\frac{2}{3}}\log(x)\,, \quad \Phi_2 = 0\,,$$
$$A_1 = A_2 = q_1(x^{-1} - x_0^{-1})L\mathrm{d}\phi\,, \qquad A_3 = q_2(x - x_0)L\mathrm{d}\phi$$

Integration constants: (η, q_1, q_2).

Two inequivalent configurations

Let us analize the range of the coordinates $[t, x, \phi, y, z]$

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- The sign of η determines whether $x > 1$ or $x < 1$. These two spacetimes are note diffeomorphic.

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- There is an asymptotic region at $x \to 1$
- The sign of η determines whether $x > 1$ or $x < 1$. These two spacetimes are note diffeomorphic.
- The spacetime closed smoothly when $\exists x_0 : F(x_0) = 0$ and $\delta = \delta(q_1, q_2, \eta)$, $\phi \in [0, \delta]$ with $\phi \sim \phi + \delta$ then

$$\eta > 0 \implies x \in]x_0, 1] \qquad \Phi_1(x) < 0$$
$$\eta < 0 \implies x \in]1, x_0] \qquad \Phi_1(x) > 0$$

Connecting with AdS

The asymptotic region is at $x \to 1$ then we need to require that

$$\Omega(x) \overset{x\to 1}{=} \frac{\rho^2}{L^2} \qquad \Longleftrightarrow \qquad \frac{x^{3/2}\eta}{(x-1)} \overset{x\to 1}{=} \frac{\rho^2}{L^2}\,,$$

This implies that

$$x = 1 + \frac{L^2\eta}{\rho^2} + \frac{2L^4\eta^2}{3\rho^4} + o(\rho^{-8})$$

Leading to $\quad \mathrm{d}s_5^2 \to \frac{\rho^2}{L^2}(-\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}\phi^2) + \frac{\mathrm{d}\rho^2}{\rho^2}$

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The dilaton $\Phi_1 = \sqrt{\frac{2}{3}}\log x$ goes to infinity as follows

$$\Phi_1 \to \sqrt{\frac{2}{3}}\frac{L^2\phi}{\rho^2} + \frac{L^4\eta^2}{3\sqrt{6}\rho^4} + o(\rho^{-8})\,.$$

The solitons excite a VEV in the dual theory of conformal dimension $\Delta = 2$

Fast review on holographic 1-point function

In an AlAdS, given a classical action

$$S[\Phi, A, G_{\mu\nu}]$$

We can compute the renormalized exact 1-point functions, by deriving the renormalized on-shell action w.r.t. to the sources

$$\Phi \to \langle O(x) \rangle_s = \frac{1}{\sqrt{g_{(0)}(x)}} \frac{\delta S_{\mathrm{ren}}}{\delta \phi_{(0)}(x)} \,,$$

$$A_\mu \to \langle J_i(x) \rangle_s = \frac{1}{\sqrt{g_{(0)}(x)}} \frac{\delta S_{\mathrm{ren}}}{\delta A_{i(0)}(x)} \,,$$

$$G_{\mu\nu} \to \langle T_{ij}(x) \rangle_s = \frac{2}{\sqrt{g_{(0)}(x)}} \frac{\delta S_{\mathrm{ren}}}{\delta g_{ij(0)}(x)} \,,$$

where $\phi_{(0)}, A_{i(0)}, g_{ij(0)}$ are the sources.

Fast review: Scalar on AdS

In the standard case $2\Delta \neq d$ the expansion of the scalar Φ solving

$$\frac{1}{\sqrt{-g}}(\sqrt{-g}g^{\mu\nu}\partial_\nu\Phi) - m^2\Phi = 0$$

in an AlAdS space $z = L^2/\rho$ with radius L and we look for solutions of the form

$$\Phi(x,z) = z^{d-\Delta}\phi(x,z)$$
$$\phi(x,z) = \phi_{(0)}(x) + z^2\phi_{(2)}(x) + z^4\phi_{(4)}(x) + \dots$$

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We get an equation of the form

$$(L^2m^2 - \Delta(\Delta - d))\phi(x,z)$$
$$-z[L^{-2}z\Box_{\text{flat}}\phi(x,z) + (d - 2\Delta + 2)\frac{\partial}{\partial z}\phi(x,z) + z\frac{\partial^2}{\partial z^2}\phi(x,z)] = 0$$

Fast review: Scalar on AdS

We solve order by order

$$\phi_{(2)}(x) = \frac{1}{2(2\Delta - d - 2)} \Box_{\text{flat}} \phi_{(0)}(x) \,,$$

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$$\vdots$$

$$\phi_{(2n)}(x) = \frac{1}{2(2\Delta - d - 2n)} \Box_{\text{flat}} \phi_{(2n-2)}(x)$$

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$$\vdots$$

$$\phi_{(2n)}(x) = \frac{1}{2(2\Delta - d - 2n)} \Box_{\text{flat}} \phi_{(2n-2)}(x)$$

The expansion is not good for the term $\phi_{(2\Delta-d)}$ it does not solve the equations

$$\Phi(x,z) = z^{d-\Delta}[\phi_{(0)}(x) + z^2\phi_{(2)}(x) + \cdots + z^{2\Delta-d-2}\phi_{(2\Delta-d-2)}(x) + z^{2\Delta-d}(\log z^2 \tilde{\phi}_{(2\Delta-d)} + \phi_{(2\Delta-d)}) + \ldots]$$

The log term is solved as

$$\tilde{\phi}_{(2\Delta-d)} = -\frac{1}{2^{2\Delta-d}\Gamma(\Delta-\frac{d}{2})(\Delta-\frac{d-2}{2})}(\Box_{\text{flat}})^{\Delta-\frac{d}{2}}\phi_{(0)}\,,$$

- All the coefficients $\phi_{(2)}, \phi_{(4)}, \ldots, \phi_{(2\Delta-d-2)}, \tilde{\phi}_{(2\Delta-d)}$ are fixed by the source $\phi_{(0)}$.
- While $\phi_{(2\Delta-d)}$ is not fixed by the near-boundary analysis.

$$\begin{aligned}\langle O\rangle_s &= \frac{1}{\sqrt{-g_{(0)}}}\frac{\delta S_{\text{ren}}}{\delta f_{(0)}} \\ &= -(2\Delta-d)\phi_{(2\Delta-d)} + F(\phi_0)\end{aligned}$$

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In our case, $d=4$ and the expansion the scalar is

$$\Phi \propto z^2 + \cdots \implies \Delta = 2$$

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In our case, $d=4$ and the expansion the scalar is

$$\Phi \propto z^2 + \cdots \implies \Delta = 2$$

$\implies$ The above expansion doesn't apply!

- $2\Delta - d = 0 \implies$ BF bound is saturated $m^2L^2 \geq -\frac{d^2}{4}$.
- In our case $d = 4, \Delta = 2$ the expansion of the scalar the expansion is

source VEV

$$\Phi = z^2 \log z^2 (\phi_{(0)} + z^2 \phi_{(2)} + \dots) + z^2 (\tilde{\phi}_0 + z^2 \tilde{\phi}_2 + \dots),$$

- In our case the source is zero, then there is only VEV $\Phi_0 \propto \eta$ with ['01 Bianchi,Freedman,Skenderis]

$$\langle O \rangle = 2\Phi_0 = 2\sqrt{\frac{2}{3}} L^2 \eta \,.$$

The solitons are a generalization of the Coulomb branch where η interpolates between the two known cases and q_1, q_2 are new sources in the dual theory.

The space of solutions

Space of solutions

A soliton solution is fully characterized in terms of its boundary conditions.
So far we use the parameters (η, q_1, q_2) defining $x_0 = x_0(\eta, q_1, q_2)$ and $\delta = \delta(\eta, q_1, q_2)$.
The physical variables:

- The bondary values of the gauge fields

$$\lim_{x\to 1} A_1 = \lim_{x\to 1} A_2 \,, \qquad \lim_{x\to 1} A_3$$

- The periodicity δ of the soliton S^1

$$\phi \sim \phi + \delta$$

Appropriate variables

- Periodicity of the boundary compact coordinate:

The existence of x_0 s.t. $F(x_0) = 0$ allows us to smoothly closed the spacetime at $x = x_0$ if the $\phi \sim \phi + \delta$ where

$$\delta = \frac{(x_0 - 1)^{1/2} x_0 F'(x_0)}{2\pi} = \frac{2\pi x_0^2}{L|q_2^2 x_0^2 + q_2 x_0 - 2q_1^2|} \sqrt{\frac{\eta}{x_0 - 1}} \, .$$

Appropriate variables

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Recall that
$\mathrm{d}s_5^2 = \Omega(x) F(x) \mathrm{d}\phi^2 + \ldots$

- Boundary value of the gauge fields:

It is convenient to parameterize the system in terms of a normalized 5-dimesnsion "Wilson lines"

$$\psi_1 \equiv \lim_{x \to 0} \frac{1}{2\pi L} \oint A^{1,2} \,, \qquad \psi_2 \equiv \lim_{x \to 0} \frac{1}{2\pi L} \oint A^3$$

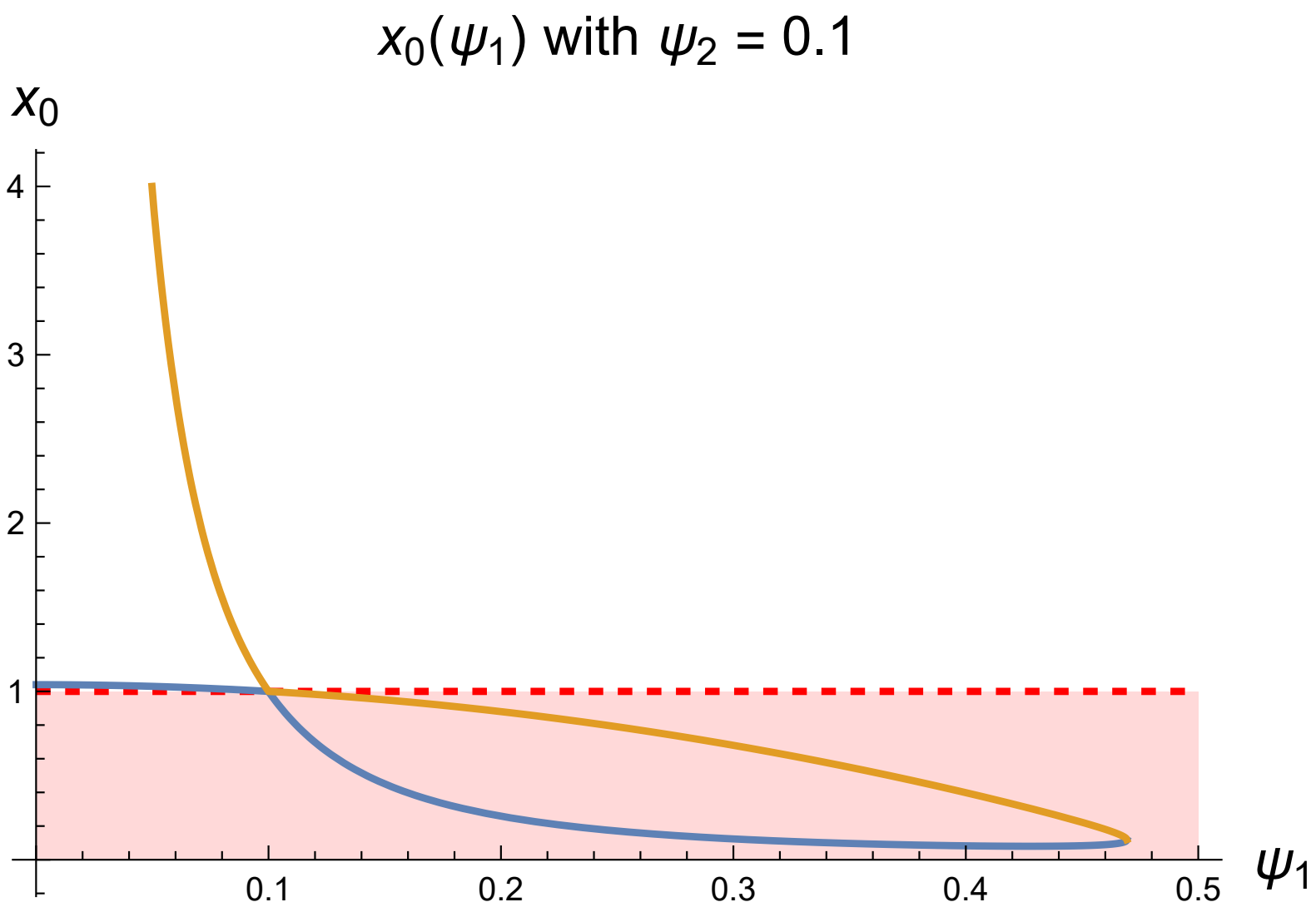
$$\Longrightarrow \qquad q_1 = \frac{2\pi x_0 \psi_1}{(x_0 - 1)\delta} \,, \qquad q_2 = \frac{2\pi \psi_2}{(1 - x_0)\delta} \,,$$

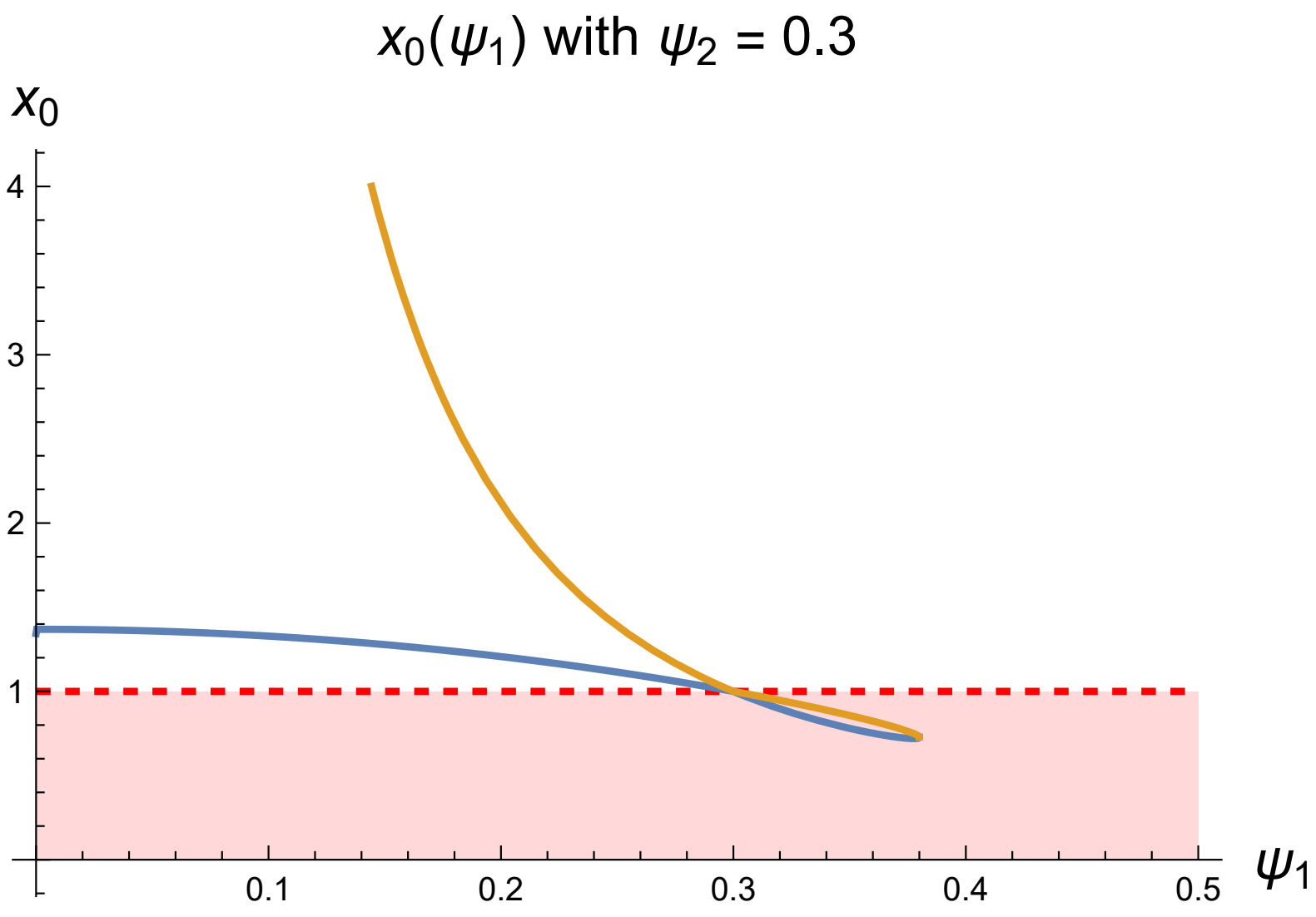
Appropriate variables

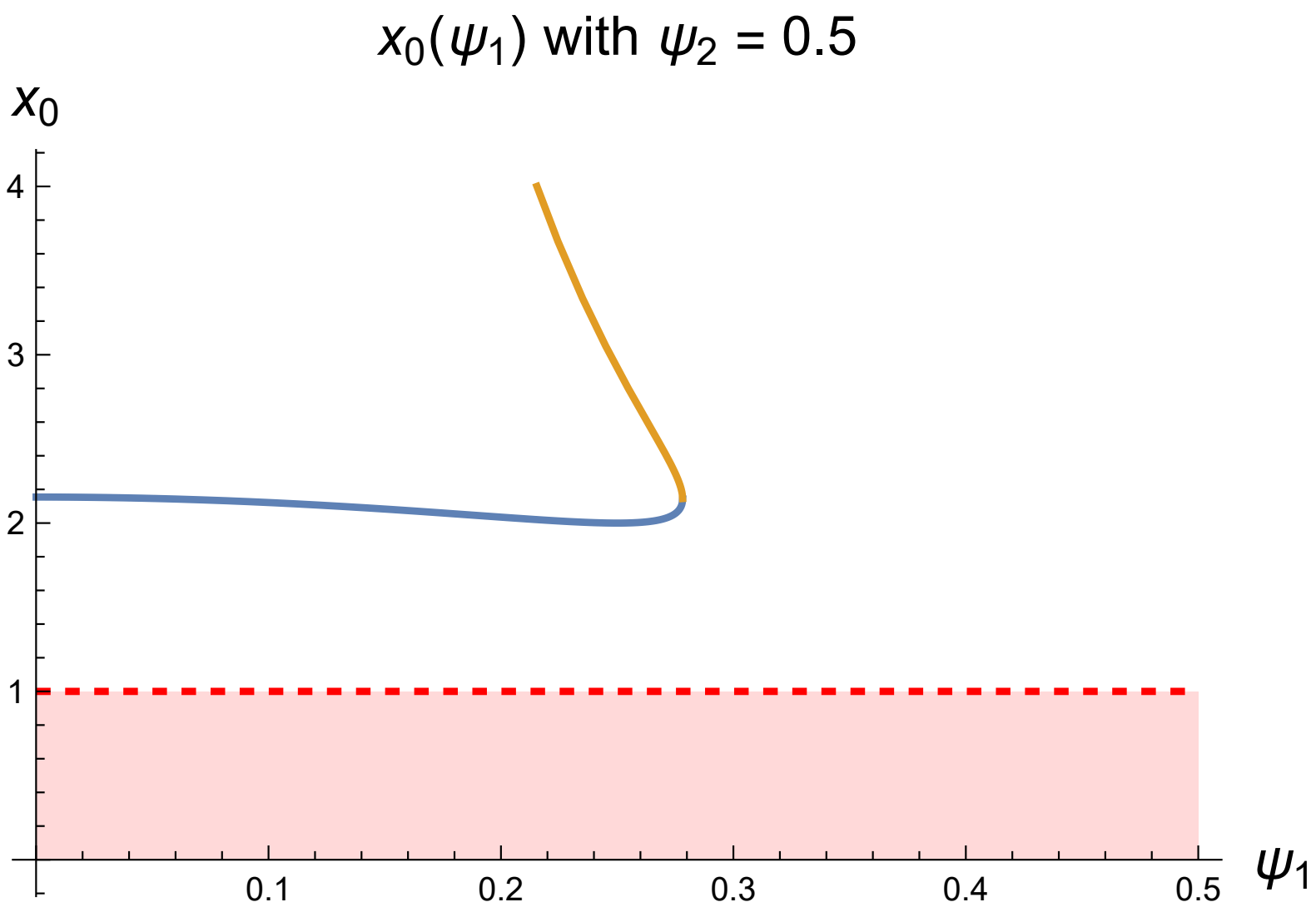
Combing the definition of $\delta(q_1, q_2, \eta; x_0)$ with $\psi_1(x_0, \delta, q_1)$ and $\psi_2(x_0, \delta, q_2)$ we get an equation for x_0

$$\begin{aligned}\psi_1^2 x_0^3 + (\psi_2^4 + 4\psi_1^4 - 4\psi_2^2\psi_1^2 - \psi_2^2 - \psi_1^2)x_0^2 \\ -\psi_2^2(4\psi_1^2 - 2\psi_2^2 - 1)x_0 + \psi_2^4 = 0\end{aligned}$$

- There are 2 sensible solutions of x_0.
- Therefore all the old variables are written in terms of (ψ_1, ψ_2, δ)
- With this parametrization we can cover $\eta < 0$ and $\eta > 0$.


$x_0(\psi_1)$ with $\psi_2 = 0.1$
x_0
4
3
2
1
0.1
0.2
0.3
0.4
0.5
ψ_1


$x_0(\psi_1)$ with $\psi_2 = 0.3$
x_0
4
3
2
1
0.1
0.2
0.3
0.4
0.5
ψ_1


$x_0(\psi_1)$ with $\psi_2 = 0.5$
x_0
4
3
2
1
0.1
0.2
0.3
0.4
0.5
ψ_1

Holographic renormalization

Holographic renormalization

The renormalized action for our situation is [2001 Bianchi,Freedman,Skenderis]

$$S_{\text{subs}} = S_0 + \frac{1}{\kappa}\int_{\partial\mathscr{M}_5} \mathrm{d}^4x\sqrt{-h}K + \frac{1}{2\kappa}\int_{\partial\mathscr{M}_5} \mathrm{d}^4x\sqrt{-h}\left[-\frac{6}{L} + \frac{1}{2L}\left(\frac{1}{\log(\rho/\rho_0)} - 2\right)\Phi_1^2\right]$$

- S_0 is the action of the 5 dimension gauged sugra
- $g_{\mu\nu} = h_{\mu\nu} + N_\mu N_\nu$
- N_μ is the outwards pointing normal to $\partial\mathscr{M}_5$.
- K is the trace of the extrinsic curvature $K_{\mu\nu} = \nabla_{(\mu}N_{\nu)}$

Holographic renormalization

The boundary metric is

$$\mathrm{d}s_4^2 = \gamma_{ab}\mathrm{d}x^a\mathrm{d}x^b = -\mathrm{d}t^2 + \mathrm{d}y^2 + \mathrm{d}z^2 + \mathrm{d}\phi^2$$

The VEV of the of the energy momentum tensor of the dual field theory is

$$\begin{aligned}\langle T_{ab}\rangle &= \frac{-2}{\sqrt{-\gamma}}\frac{\delta S_{\mathrm{ren}}}{\delta\gamma^{ab}} = \lim_{\rho\to\infty}\frac{\rho^2}{L^2}\frac{-2}{\sqrt{-h}}\frac{\delta S_{\mathrm{subs}}}{\delta h^{ab}}\,,\\ &= \lim_{\rho\to\infty}\frac{\rho^2}{L^2\kappa}\left(h_{ab}K - K_{ab} - \frac{3}{L}h_{ab} - \frac{1}{2L}h_{ab}\Phi_1^2\right)\end{aligned}$$

replacing the fields and taking the limits,

$$\langle T_{tt}\rangle = \frac{\eta L(q_1^2 - q_2^2)}{2\kappa}\,,$$

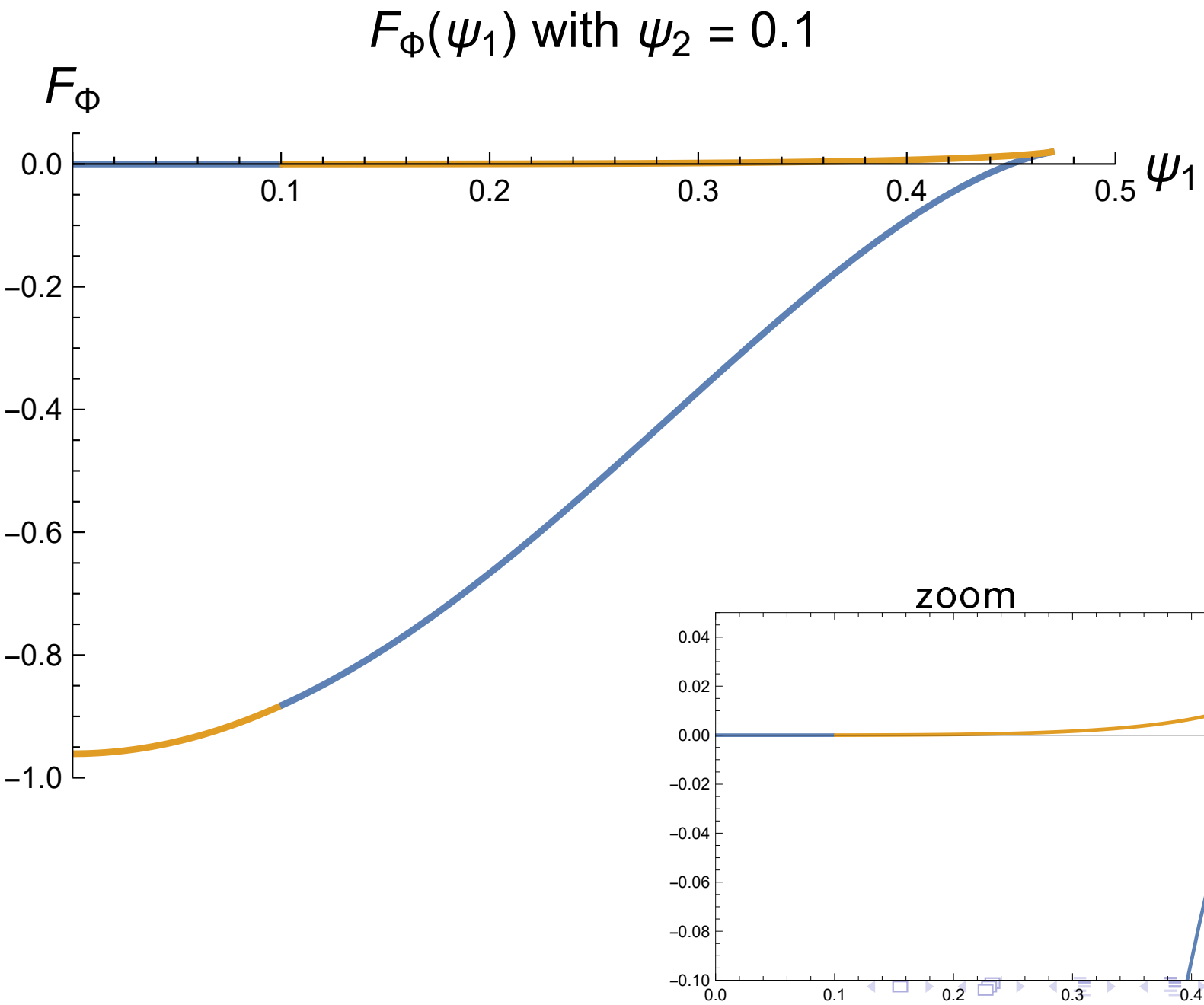
$$\langle T_{yy}\rangle = \langle T_{zz}\rangle = -\langle T_{tt}\rangle\,, \quad \langle T_{\phi\phi}\rangle = -\frac{3}{2}\langle T_{tt}\rangle$$

Phase diagram

Free energy in the canonical ensemble is the energy $E_\Phi \equiv \langle T_{tt} \rangle V_2 \delta$

$$F_\Phi \equiv \frac{E_\Phi}{|E_0|} = -16 \frac{(\psi_1^2 x_0^2 - \psi_2^2)(\psi_1^2 x_0 - \psi_2^2)}{x_0 (x_0 - 1)^2}$$

where we are normalizing it w.r.t. the AdS soltion energy ['98 Horowitz,Myers] $E_0 = -\frac{L^3 \pi^4}{2\kappa\delta^3} V_2$


$F_\Phi(\psi_1)$ with $\psi_2 = 0.1$
F_Φ
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.1
0.2
0.3
0.4
0.5
ψ_1
zoom
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
0.0
0.1
0.2
0.3
0.4
0.5

Phase transition

- Increasing ψ_1 until $|\psi_1| = |\psi_2|$ the scalar changes sing and a re-distribution of the D3-brane happens.
- Phase transition at $|\psi_1| = |\psi_2|$.
- $F_\Phi = 0$ corresponds to the energy of ['98 Freedman, Gubser, Pilch, Warner].

QFT energy

Toy model: Consider a QFT in Minkowski 4 with $\phi \sim \phi + \delta$

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}y^2 + \mathrm{d}z^2 + \mathrm{d}\phi^2$$

the Casimir energy of a scalar field is

$$\langle E_{\mathrm{QFT}} \rangle = -\frac{\pi^2}{6\delta^3} V_2 X$$

X depends on the periodicity of the field in the S^1

$$X_{\mathrm{perdiodic}} = \sum_{n=1}^{\infty} (2n)^3 = \frac{8}{120}\,,$$

$$X_{\mathrm{anti-periodic}} = \sum_{n=1}^{\infty} (2n-1)^3 = -\frac{7}{120}\,.$$

QFT energy

If we apply this calculation to $\mathcal{N} = 4$ SYM at weak coupling we find that

- Fermions: anti-periodic, Bosons: anti-periodic

$$\langle E_{\mathrm{QFT}} \rangle = -\frac{\pi^2 V_2}{6\delta^3} X_{\mathrm{periodic}} (N^2 - 1)(6 + 2 - 4 \times 2) = 0\,,$$

QFT energy

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- Fermions: anti-periodic, Bosons: periodic

$$\begin{aligned} \langle E_{\mathrm{QFT}} \rangle_{\mathrm{mixed}} &= -\frac{\pi^2 V_2}{6\delta^3} X_{\mathrm{periodic}} (N^2 - 1)(6 + 2 + 8\frac{7}{8}) \\ &= -\frac{\pi^2 V_2}{6\delta^3} (N^2 - 1)\,, \end{aligned}$$

in the large N and considering $\frac{L^3}{\kappa} = \frac{N^2}{4\pi^2}$ we find that

$$E_0 = \frac{3}{4} \langle E_{\mathrm{QFT}} \rangle_{\mathrm{mixed}}$$

Lift to 10 dimensions

Using the lift in the deformed S^5 we show that our configuration is a solution of IIB sugra.
Change of variable

$$x = \frac{1}{1 + \epsilon \frac{\ell^2}{r^2}}$$

where $\eta = -\epsilon \ell^2 / L^2$ and $\epsilon = \pm 1$. The relevant functions are

$$F(r) \equiv L^{-2} - \frac{\epsilon \ell^2 L^2}{r^4} \left(q_1^2 - q_2^2 \lambda(r)^{-6} \right)$$

$$\lambda(r)^6 \equiv 1 + \epsilon \frac{\ell^2}{r^2}$$

$$\zeta(r, \theta)^2 = 1 + \epsilon \frac{\ell^2}{r^2} \cos\theta$$

Lift to 10 dimensions

The metric in 10 dimensions in coordinates $[x_{1,2}, r, \phi, \theta, \phi_3, \psi, \phi_1, \phi_2]$

$$
\begin{aligned}
&\mathrm{d}s_{10}^2 = \\
&\frac{\zeta(r,\theta) r^2}{L^2}\left(\frac{L^2 \mathrm{d}r^2}{r^4 F(r)\lambda(r)^6} + \mathrm{d}x_{1,2}^2 + F(r) L^2 \mathrm{d}\phi^2\right) \\
&+ \frac{L^2}{\zeta(r,\theta)}\left\{\zeta(r,\theta)^2 \mathrm{d}\theta^2 + \lambda(r)^6 \sin^2\theta \left(\mathrm{d}\phi_3 + L^{-1} A_3\right)^2\right. \\
&\left. + \cos^2\theta \left[\mathrm{d}\psi^2 + \sin^2\psi \left(\mathrm{d}\phi_1 + L^{-1} A_1\right)^2 + \cos^2\psi \left(\mathrm{d}\phi_2 + L^{-1} A_2\right)^2\right]\right\}
\end{aligned}
$$

Lift to 10 dimensions

The F_5 is given by

$$F_5 = G_5 + \star G_5$$

and $G_5 = \mathrm{d}\mathcal{C}_4$

$$\mathcal{C}_4 = -\left[\frac{r^4}{L^4}\zeta(r,\theta)^2 + \frac{\ell^4}{r_0^2}\epsilon\cos^2\theta(q_2^2\lambda(r_0)^{-6} - q_1)\right]\mathrm{d}t\wedge\mathrm{d}y\wedge\mathrm{d}z\wedge\mathrm{d}\phi$$
$$+\ell^2\epsilon[q_1\cos^2\theta(\cos^2\psi\mathrm{d}\phi_2 + \sin^2\psi\mathrm{d}\phi_1)+$$
$$+q_2\cos^2\theta\mathrm{d}\phi_3]\wedge\mathrm{d}t\wedge\mathrm{d}y\wedge\mathrm{d}z$$

The flux of the F_5 on the S^5 with coord. $[\theta,\phi_3,\psi,\phi_1,\phi_2]$ with $\theta,\psi\in[0,\pi/2]$ and $\phi_1,\phi_2,\phi_3\in[0,2\pi]$ is

$$\int_{S^5} F_5 = \int_{S^5}\star F_5 = 4\pi^3 L^4\epsilon$$

BPS configurations (detail)

To check whether a spacetime perserves some amount of supersymmetry we need to solve

$$\delta\lambda \equiv \mathbf{T}\epsilon = 0 \ , \qquad \delta\psi_\mu dx^\mu \equiv \mathbf{D}\epsilon = d\epsilon + \mathbf{W}\epsilon = 0 \ ,$$

for killing spinor ϵ. The strategy is

1. Compute the integrability condition for the Killing spinor

$$\mathbf{D} \wedge \mathbf{D}\epsilon = 0 \qquad \rightarrow \qquad \underbrace{(d\mathbf{W} + \mathbf{W} \wedge \mathbf{W})}_{\Xi_{\mu\nu}} \epsilon = 0$$

2. Solve the determiant equations

$$\det \Xi_{\mu\nu} = 0 \qquad \wedge \qquad \det \mathbf{T} = 0$$

   which imposes restrictions on the parameters of the background.

3. Finaly solve the linear equation

$$d\epsilon + \mathbf{W}\epsilon = 0$$

In $D = 5$ BPS configuration

In $D = 5$, we have that the gravitino matrix

$$\mathbf{W} \equiv \frac{1}{4}\omega_{ab}\gamma^{ab} + L^{-1}\sum_i(-\frac{\mathrm{i}}{2}A^i + \frac{1}{3!}\gamma_c e^c X_i)$$
$$+ \frac{\mathrm{i}}{4!}(\gamma_c\gamma^{ab} - 6\delta^a_c\gamma^b)e^c\sum_i X_i^{-1}F^i_{ab}$$

and the dilatinos matrices

$$\mathbf{\Omega}_i \equiv -\frac{1}{8}X_i^{-2}\gamma^{ab}F^i_{ab} - \frac{\mathrm{i}}{4}X_i^{-2}\left(\frac{\partial X_i}{\partial\Phi_1}\partial\!\!\!/\Phi_1 + \frac{\partial X_i}{\partial\Phi_2}\partial\!\!\!/\Phi_2\right)\mathbb{1} + \frac{\mathrm{i}}{2L}\mathbb{1}$$

Integrability conditions

$$\det\Omega_1 = 0\,, \qquad \det\Omega_2 = 0\,, \qquad \det(\mathrm{d}\mathbf{W} + \mathbf{W}\wedge\mathbf{W})_{\mu\nu} = 0$$

hold iff

$$q_1 = -q_2$$

In $D = 10$ the BPS configuration

The only non-trivial spinor variation is the gravitino with matrix 64×64 matrix

$$\mathbf{W}_{10} = \frac{1}{4}\omega_{ab}\Gamma^{ab} + \frac{1}{16}\not{F}_5 i\sigma_2\Gamma_a e^a$$

and the IC is

$$\begin{aligned}\Xi &\equiv \mathrm{d}\mathbf{W}_{10} + \mathbf{W}_{10} \wedge \mathbf{W}_{10} \\ &= \frac{1}{4}R_{ab}\Gamma^{ab} + \frac{1}{16}\frac{1}{5!}i\sigma_2\mathcal{D}F_{b_1\ldots b_5}\Gamma^{b_1\ldots b_5} \wedge \Gamma_a e^a \\ &- \frac{1}{128}\frac{1}{4!}\not{F}_5 F_{ad_1\ldots d_4}\Gamma^{d_1\ldots d_4} e^a \wedge \Gamma_c e^c\end{aligned}$$

Then $\det \Xi_{\mu\nu} = 0$ implies that

$$|q_1| = |q_2|$$

note only $q_1 = -q_2$!!

Killing spinors in $D = 5$

In the coordiante $x = (1 + \epsilon\ell^2/r^2)^{-1}$ when $q_1 = -q_2$ there are two complex killing spinors $\hat{\epsilon} = \epsilon^1 + \mathrm{i}\epsilon^2$

$$\Psi_1 = e^{-\frac{\mathrm{i}\pi\phi}{\delta}+\sigma(r)} \begin{pmatrix} 1 \\ \Lambda(r) \\ 0 \\ 0 \end{pmatrix}, \qquad \Psi_2 = e^{-\mathrm{i}\frac{\pi\phi}{\delta}+\sigma(r)} \begin{pmatrix} 0 \\ 0 \\ 1 \\ -\Lambda(r) \end{pmatrix}$$

- They are anti-periodic in $\phi \in [0, \delta]$
- The solution is 1/8 BPS.
- In this conventions AdS_5 has 4 KS spinors
- In $D = 10$ was not possible construct the KS due to technical difficulties.

Conclusions

- We found AdS solitions depending on three parameters: two sources associated to gauge fields and the value of the circle S^1.
- In the (E, ψ_1, ψ_2) we found phase transitions between the various D3-brane distributions.
- The 10-dimensional uplift show that the solutions are deformations of the D3-brane distribution of the ['98 Freedman, Gubser, Pilch, Warner] leading a completely smooth configuration of type IIB.
- We found that the phase diagram of these solutions should correspond to the strongly coupled description of two possible vacua of the large N $\mathcal{N} = 4$ SYM.

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Thank you!